

(a) Describe an experiment to verify the inverse square law

✔ 2 marks

✔ Sample response:

- Place a **light source** (e.g. lamp or LED) at a fixed point.
- Use a **light sensor** connected to a **lux meter** or voltmeter to measure irradiance at various **measured distances** from the source.
- Ensure the room is **darkened** to reduce background light.
- Plot a graph of **irradiance vs 1/d²**. A straight-line graph confirms the inverse square law.

(b)(i) State what is meant by the term irradiance

✔ 1 mark

✔ Answer:

Irradiance is the **power incident per unit area** on a surface.

(b)(ii) Use all the data to establish the relationship between irradiance I and distance d

✔ 3 marks

Distance <i>d</i> (m)	Irradiance <i>I</i> (W/m²)	$1/d^2$ (m ⁻²)
0.200	142.0	25.0
0.300	63.1	11.1
0.400	35.5	6.25
0.500	22.7	4.00
0.600	15.8	2.78

✔ A graph of *I* vs $1/d^2$ will show a **straight line** → confirms:

$I \propto \frac{1}{d^2}$

✔ Answer:

The graph shows that irradiance is **inversely proportional to the square of the distance**.

(c) Explain why irradiance decreases with increasing distance from the point source

✔ 2 marks

✔ Answer:

As distance increases, light spreads out over a **larger area** (spherical surface), so the **power per unit area decreases**.
→ The **same total power** is distributed across a larger surface.

🧠 Revision Tips

- Inverse square law applies to **point sources** like light, gravity, and sound
- Key formula:
 $I = \frac{P}{4\pi d^2}$
- Always plot *I* vs $1/d^2$ to test the relationship
- Be consistent with units (e.g. m, W/m²)